

To show a set is a subring we need to show that it is a subgroup under addition and closed under multiplication. $a - b$ is in the subring ab is in the subring.

Section 7.3

Exercise (7.3.10). Decide which of the following are ideals of the ring $\mathbb{Z}[x]$

- b. The set of all polynomials whose coefficient of x^2 is a multiple of 3.

Proof: Let G be the set of all polynomials whose coefficient of x^2 is a multiple of 3.

Note that the polynomials $p(x) = 0x^2 + x$ and $q(x) = 3x^2 + x$ are in G . By multiplication we find that $p(x)q(x) = 3x^3 + (1)x^2$ cannot be in G since 1 is not a multiple of 3. Thus G is not a subring of $\mathbb{Z}[x]$ and therefore not an ideal $\mathbb{Z}[x]$. $\square$

- c. The set of all polynomials whose constant term, coefficient of x and coefficient of x^2 are all zero.

Proof: Let G be the set of all polynomials whose constant term, coefficient of x and coefficient of x^2 are all zero. Let $p(x), q(x) \in G$, such that $p(x) = \alpha_3x^3 + \alpha_4x^4 \dots \alpha_jx^j$ and $q(x) = \beta_3x^3 + \beta_4x^4 \dots \beta_nx^n$ where $\alpha_0, \alpha_1, \alpha_2, \beta_0, \beta_1, \beta_2 = 0$. By polynomial addition we know that the additive inverse of $q(x) \in G$. Note that $p(x) - q(x)$ is a polynomial of the form $p(x) - q(x) = (\alpha_3 - \beta_3)x^3 + (\alpha_4 - \beta_4)x^4 \dots$, so by definition $p(x) - q(x) \in G$. Now consider $p(x)q(x)$,

$$p(x)q(x) = \sum_{i=0}^{j+n} \left(\sum_{k=0}^i \alpha_k \beta_{i-k} \right) x^i.$$

Note that the sum $\sum_{k=0}^i \alpha_k \beta_{i-k}$ is guaranteed to have zero terms until at least $i = 6$ and we get $\alpha_3\beta_3x^6$. Therefore $p(x)q(x) \in G$. Therefore G is a subring of $\mathbb{Z}[x]$.

Now let $q(x) \in \mathbb{Z}[x]$, defined as before with $\beta_0, \beta_1, \beta_2$ possibly nonzero and consider $p(x)q(x)$. Note that again, the sum $\sum_{k=0}^i \alpha_k \beta_{i-k}$ is guaranteed to have zero terms until at least $i = 3$ where we get $\alpha_3\beta_0x^3$. So $p(x)q(x) \in G$ and since multiplication is commutative in $\mathbb{Z}$ we get $q(x)p(x) \in G$. Thus G is an ideal. $\square$

- d. $\mathbb{Z}[x^2]$ (i.e., the set of all polynomials in which only even powers of x appear).

Proof: Suppose for the sake of contradiction that $\mathbb{Z}[x^2]$ is an ideal of $\mathbb{Z}[x]$. Note that

$$p(x) = x^2 \in \mathbb{Z}[x^2] \text{ and } q(x) = x \in \mathbb{Z}[x]. \text{ Since } \mathbb{Z}[x^2] \text{ is an ideal we know that } p(x)q(x) \in \mathbb{Z}[x^2] \text{ however } p(x)q(x) = x^3 \text{ which is not in } \mathbb{Z}[x^2]. \quad \square$$

Exercise (7.3.17). Let R and S be nonzero rings with identity and denote their respective identities by 1_R and 1_S . Let $\varphi : R \rightarrow S$ be a nonzero homomorphism of rings.

- a. Prove that if $\varphi(1_R) \neq 1_S$ then $\varphi(1_R)$ is a zero divisor in S . Deduce that if S is an integral domain then every ring homomorphism from R to S sends the identity of R to the identity of S .

Proof: Suppose R and S be nonzero rings, $\varphi : R \rightarrow S$ be a nonzero homomorphism of rings and $\varphi(1_R) \neq 1_S$. Let $\varphi(1_R) = s \in S$. Since φ is a homomorphism we know that $\varphi(1_R) = \varphi(1_R 1_R) = \varphi(1_R)\varphi(1_R)$, by substitution we get $s = s^2$, by subtraction we get $s - s^2 = 0$ which implies $s(1_S - s) = 0$. Since $s = \varphi(1_R) \neq 1_S$ it must follow that $(1_S - s)$ is nonzero and since φ is a nonzero homomorphism and with $s \neq 0$ we conclude that s is a zero divisor.

Suppose S was an integral domain. By definition S has no zero divisors, therefore by the contrapositive of the previous result we find that $\varphi(1_R) = 1_S$.

If $\varphi(1_R) \neq 1_S$ then S is not an integral domain. $\rightarrow$ If S is an integral domain then $\varphi(1_R) = 1_S$. $\square$

- b. Prove that if $\varphi(1_R) = 1_S$ then $\varphi(u)$ is a unit in S and that $\varphi(u^{-1}) = \varphi(u)^{-1}$ for each unit u in R .

Proof: Suppose $\varphi(1_R) = 1_S$. Let $u \in R$ be a unit, and note that $\varphi(u)\varphi(u^{-1}) = \varphi(uu^{-1}) = \varphi(1_R) = 1_S$. Thus $\varphi(u)$ is unit in S . Thus $\varphi(u^{-1}) = \varphi(u)^{-1}$. $\square$

Exercise (7.3.24). Let $\varphi : R \rightarrow S$ be a ring homomorphism.

- a. Prove that if J is an ideal of S then $\varphi^{-1}(J)$ is an ideal of R . Apply this to the special case when R is a subring of S and φ is the inclusion homomorphism to deduce that if J is an ideal of S then $J \cap R$ is an ideal of R .

Proof: Suppose J is an ideal of S and let $\varphi^{-1}(J)$ be the pre-image of J in R . We

will show that $\varphi^{-1}(J)$ is closed under subtraction and multiplication. Let $a, b \in \varphi^{-1}(J)$, and note that $\varphi(a - b) = \varphi(a) - \varphi(b)$, since J is a subring we know that $\varphi(a - b) \in J$ thus $a - b \in \varphi^{-1}(J)$. Similarly we get $\varphi(ab) = \varphi(a)\varphi(b)$, since J is a subring we know that $\varphi(ab) \in J$ thus $ab \in \varphi^{-1}(J)$.

Let $j \in J$, $\varphi^{-1}(j) \in \varphi^{-1}(J)$, and $r \in R$. Note that $\varphi(\varphi^{-1}(j)r) = \varphi(\varphi^{-1}(j))\varphi(r) = j\varphi(r) = j'$ where $j' \in J$ since J is an ideal. Since $\varphi^{-1}(j)r$ maps to an element in J it must be the case that $\varphi^{-1}(j)r \in \varphi^{-1}(J)$. Since J is an ideal the same argument can be made for $r\varphi^{-1}(j) \in \varphi^{-1}(J)$. Thus $\varphi^{-1}(J)$ is an ideal of R .

Suppose R is a subring of S and φ is the inclusion homomorphism. In this case $\varphi^{-1}(J)$, the pre-image of J in R is simply $J \cap R$ which by the previous result must be an ideal of R . □

- b. Prove that if φ is surjective and I is an ideal of R then $\varphi(I)$ is an ideal of S . Give an example where this fails if φ is not surjective.

Proof: Suppose that φ is surjective and I is an ideal of R . Let $a, b \in \varphi(I)$ and note

that there exists $i, j \in I$ such that $\varphi(i) = a$, $\varphi(j) = b$. By substitution we get, $a - b = \varphi(i) - \varphi(j) = \varphi(i - j)$, since I is also a subring we get that $i - j \in I$ so $a - b = \varphi(i - j) \in \varphi(I)$. Similarly we know that $ab = \varphi(i)\varphi(j) = \varphi(ij)$, and since I is also a subring we know that $ij \in I$ so $ab = \varphi(ij) \in \varphi(I)$.

Let $s \in S$ and note that since φ is a surjection there exists some $n \in R$ such that $\varphi(n) = s$. Note that $as = \varphi(i)\varphi(n) = \varphi(in)$ since I is an ideal we know that $in \in I$ and thus $as \in \varphi(I)$. The same argument can be made for sa since I is an ideal. Thus $\varphi(I)$ is an ideal in S .

For an example consider the inclusion map from $\mathbb{Z}$ to $\mathbb{Q}$. Let $I = \mathbb{Z}$ and note that $\varphi(\mathbb{Z}) = \mathbb{Z}$ in $\mathbb{Q}$. We find that $1(\frac{1}{3}) = \frac{1}{3} \notin I$. □

Exercise (7.3.34). Let I and J be ideal in R .

- a. Prove that $I + J$ is the smallest ideal of R containing both I and J .

Proof: Suppose I and J be ideal in R and consider $I + J = \{a + b : a \in I, b \in J\}$. Let $a, b \in I + J$ such that $a = a_i + a_j$ and $b = b_i + b_j$. Note that $a - b = (a_i + a_j) - (b_i + b_j) = (a_i - b_i) + (a_j - b_j)$ since I, J are both subrings we get that $a - b = (a_i - b_i) + (a_j - b_j) \in I + J$. Note that $ab = (a_i + a_j)(b_i + b_j) = a_i(b_i + b_j) + a_j(b_i + b_j)$ since I, J are both ideals we get that $ab = a_i(b_i + b_j) + a_j(b_i + b_j) \in I + J$. Similarly let $r \in R$ and note that $ar = (a_i + a_j)r = a_i r + a_j r$ and since I, J are both ideals we get that $ar = a_i r + a_j r \in I + J$. The same argument applies to show $ra \in I + J$.
Now suppose that S is an ideal of R such that $I, J \subseteq S$. Note that $a_i, a_j \in S$ and since S is also a subring we know that $a_i + a_j \in S$ thus $I + J \subseteq S$. $\square$

- b. Prove that IJ is an ideal contained in $I \cap J$.

Proof: Suppose I and J be ideal in R and consider IJ . Let $a, b \in IJ$, by definition we know that $a = \sum_k^n \alpha_k \beta_k$ and $b = \sum_k^m i_k j_k$ where $\alpha_k, i_k \in I$ and $\beta_k, j_k \in J$. Note that $a - b = \sum_k^n \alpha_k \beta_k - \sum_k^m i_k j_k$ is just another finite sum of products of ideals I and J . Note that $ab = (\sum_k^n \alpha_k \beta_k)(\sum_k^m i_k j_k) = \sum_k^n (\sum_l^m i_l j_l) \alpha_k \beta_k$ since I is an ideal we know that $(\sum_l^m i_l j_l) \alpha_k = O_k$ for some $O_k \in I$. Thus $ab = \sum_k^n O_k \beta_k \in IJ$.
Let $r \in R$ and note that $ar = (\sum_k^n \alpha_k \beta_k)r = \sum_k^n \alpha_k (\beta_k r)$, since J is an ideal we know that $(\beta_k r) \in J$ and thus $ar \in IJ$. A similar argument can be made for ra , noting that I is an ideal. Note that by definition $a = \sum_k^n \alpha_k \beta_k$ every term $(\alpha_k \beta_k)$ is in I and J since $\alpha_k \in I$ and $\beta_k \in J$. Therefore $a \in I \cap J$ and thus IJ is an ideal contained in $I \cap J$. $\square$

- c. Give an example where $IJ \neq I \cap J$.

Proof: Let $I = 2\mathbb{Z}$ and $J = 4\mathbb{Z}$. Note that $IJ = 2\mathbb{Z}4\mathbb{Z} = 8\mathbb{Z}$ and $I \cap J = 2\mathbb{Z} \cap 4\mathbb{Z} = 4\mathbb{Z}$. $\square$

d. Prove that if R is commutative and if $I + J = R$ then $IJ = I \cap J$.

Proof: Suppose R is commutative and $I + J = R$. By the previous results we have show that $IJ \subseteq I \cap J$. Let $a \in I \cap J$. Note that since I and J are ideals we know that $I \cap J$ is an ideal. Let $r \in R$, since $R = I + J$ let $r = i + j$. Since $I \cap J$ is an ideal $ar \in I \cap J$. Note that $ar = a(i + j) = ai + aj$. Since I and J are ideals $ar \in I + J$. Therefore $I \cap J \subseteq IJ$ thus $IJ = I \cap J$. $\square$

Section 7.4

Exercise (7.4.5). Let R be a ring with identity $1 \neq 0$. Prove that if M is an ideal such that R/M is a field then M is a maximal ideal.

Proof: Suppose M is an ideal such that R/M is a field. For the sake of contradiction suppose M is not a maximal ideal. Therefore there exists some ideal I such that $M \subseteq I \subset R$. By the Lattice Isomorphism Theorem the ideals of R containing M correspond bijectively to the ideals of R/M . Note that R/M is commutative, let $aM, bM \in R/M$ and note that $aMbM = aMM = aM = M = bM = bMM = bMaM$. Since R/M is also a field we know by Proposition 9 that it's only ideals must be 0 and R/M . Since the Lattice Isomorphism Theorem gives us three ideals of R/M but Proposition 9 gives only 2, we have a contradiction. $\square$

Exercise (7.4.11). Assume R is commutative. Let I and J be ideals of R and assume P is a prime ideal of R that contains IJ . Prove either I or J is contained in P .

Proof: Suppose I and J be ideals of R and let P be a prime ideal of R that contains IJ .

Without loss of generality suppose $I \not\subseteq P$. Let $a \in IJ$ such that $a = ij$ where $i \in I$ and $j \in J$. Note that by the definition of a prime ideal since R is commutative we

know that if $ij \in P$ either $i \in P$ or $j \in P$. Since $I \not\subseteq P$ we conclude that $j \in P$ and therefore $J \subseteq P$. $\square$

Exercise (7.4.13). Let $\varphi : R \rightarrow S$ be a homomorphism of commutative rings.

- a Prove that if P is a prime ideal of S then either $\varphi^{-1}(P) = R$ or $\varphi^{-1}(P)$ is a prime ideal of R . Apply this to the special case when R is a subring of S and φ is the inclusion homomorphism to deduce that if P is a prime ideal of S then $P \cap R$ is either R or a prime ideal of R .

Proof: Suppose P is a prime ideal of S . By 7.3.24(a) we know that $\varphi^{-1}(P)$ is an ideal in R . Consider $a, b \in R$ such that $ab \in \varphi^{-1}(P)$. Note that $\varphi(ab) = \varphi(a)\varphi(b) \in P$ and since P is a prime ideal and S is commutative at least one of the two elements $\varphi(a) \in P$ or $\varphi(b) \in P$. If only one element is in P without loss of generality we know that $a \in \varphi^{-1}(P)$ and thus $\varphi^{-1}(P)$ is prime. If both elements are in P we conclude that $R \subseteq \varphi^{-1}(P)$ and therefore $\varphi^{-1}(P) = R$ since we would conclude that both $a, b \in \varphi^{-1}(P)$.

Again like in problem 7.3.24(a) when R is a subring of S and φ is the inclusion map $\varphi^{-1}(P) = P \cap R$ and the result follows from above. $\square$

- b Prove that if M is a maximal ideal of S and φ is surjective then $\varphi^{-1}(M)$ is a maximal ideal of R . Give an example to show that this need not be the case of φ is not surjective.

Proof: Suppose M is a maximal ideal of S and φ is surjective. For the sake of contradiction suppose $\varphi^{-1}(M)$ is not a maximal ideal of R , then there exists some ideal I such that $\varphi^{-1}(M) \subset I \subset R$. By 7.3.24(b) since φ is surjective and I is an ideal it must follow that $\varphi(I)$ is an ideal in S . However since $\varphi^{-1}(M) \subset I$ and M is maximal in S it must follow that $\varphi(I) = S$. Yet since φ is surjective it must also follow that $I = R$ a contradiction. $\square$

Section 8.1

Exercise (8.1.1). For each of the following pairs of integers a and b , determine their greatest common divisor d and write d as a linear combination $ax + by$ of a and b .

a. $a = 20$, $b = 13$

Proof: Using the euclidean algorithm we get the following,

$$20 = 13(1) + 7$$

$$13 = 7(1) + 6$$

$$7 = 6(1) + 1$$

$$6 = 6(1) + 0$$

Which gives us $(20, 13) = 1$. By back substitution we get the following,

$$1 = 7 - 6 = 7 - (13 - 7) = 2(7) - 13 = 2(20 - 13) - 13 = 2(20) - 3(13). \quad \square$$

c. $a = 11391$, $b = 5673$

Proof: Using the euclidean algorithm we get the following,

$$11391 = 5673(2) + 45$$

$$5673 = 45(126) + 3$$

$$45 = 3(15) + 0$$

Which gives us $(11391, 5673) = 3$. By back substitution we get the following,

$$3 = 5673 - 45(126) = 5673 - (11391 - (2)5673)(126) = 253(5673) - 126(11391)$$

$\square$

e. $a = 91442056588823$, $b = 779086434385541$

Proof: Using the euclidean algorithm we get the following,

$$779086434385541 = 91442056588823(8) + 47549981674957$$

$$91442056588823 = 47549981674957(1) + 43892074913866$$

$$47549981674957 = 43892074913866(1) + 3657906761091$$

$$43892074913866 = 3657906761091(11) + 3655100541865$$

$$3657906761091 = 3655100541865(1) + 2806219226$$

$$3655100541865 = 2806219226(1302) + 1403109613$$

$$2806219226 = 1403109613(2) + 0$$

Which gives us $(91442056588823, 779086434385541) = 1403109613$. By back substitution we get, $1403109613 = 277522(91442056588823) - 32573(779086434385541)$

□

Exercise (8.1.2). For each of the following pairs of integers a and n , show that a is relatively prime to n and determine the inverse of $a \pmod n$

a. $a = 13, n = 20$

Proof: In the previous problem we found that a and n are relatively prime since the

$\gcd(a, n) = 1$. We also showed that $1 = 2(20) - 3(13)$ so we get $-3(13) \cong 1 \pmod{20}$ and therefore $-3 \pmod{20} = 17 \pmod{20}$ is the inverse of 13. □

c. $a = 1891, n = 3797$

Proof: By the euclidean algorithm we get the following,

$$3797 = 1891(2) + 15$$

$$1891 = 15(126) + 1$$

$$15 = 1(15) + 0.$$

Which gives us that $(1891, 3797) = 1$. By back substitution we get that $1 = 253(1891) - 126(3797)$. Therefore we know that $-126(3797) = 1 \pmod{1891}$ and therefore $-126 \pmod{1891} = 1765 \pmod{1891}$ is the inverse of $3797 \pmod{1891}$.

□

Exercise (8.1.3). Let R be a Euclidean Domain. Let m be the minimum integer in the set of norms of nonzero elements of R . Prove that every nonzero element of R of norm m is a unit. Deduce that a nonzero element of norm zero (if such an element exists) is a unit.

Proof: Let R be a Euclidean Domain. Suppose m is the minimum integer in the set of norms of nonzero elements of R . Choose $n \in R$ such that $N(n) = m$. By definition of Euclidean Domain for all $a \in R$ there exists $q, r \in R$ such that $a = qn + r$ where either $r = 0$ or $N(r) < N(n)$. Since $N(n)$ is minimal for all R we know that $r = 0$ and therefore $a = qn$. Since $1 \in R$ there exists a $q \in R$ such that $1 = qn$ thus n is a unit in R .

Since the norm function is defined on $N : R \rightarrow \mathbb{Z} \cup \{0\}$ we know that if a $n \in R$ such that $N(n) = 0$ and $n \neq 0$ will have the property that for every $a \in R$ there exists $q, r \in R$ such that $a = qn + r$ forces $r = 0$. Since $1 \in R$ we know that for some q , $1 = qn$. Thus n is a unit. □

Exercise (8.1.7). Find the generator for the ideal $(85, 1 + 13i)$ in $\mathbb{Z}[i]$, i.e. the greatest common divisor for 85 and $1 + 13i$, by the Euclidean Algorithm. Do the same for the ideal $(47 - 13i, 53 + 56i)$.

Solution: As outlined in Example 3 of Section 8.1 we know that The Gaussian integers are a Euclidean Domain under the norm $N(a + bi) = a^2 + b^2$. Recall the division algorithm for The Gaussian integers under norm N , for all $\alpha, \beta \in \mathbb{Z}[i]$ there exists some $(p + qi), \gamma \in \mathbb{Z}[i]$ such that $\alpha = (p + qi)\beta + \gamma$ where $N(\gamma) \leq \frac{1}{2}N(\beta)$. We begin

the Euclidean algorithm, note that

$$\frac{85}{1+13i} = \frac{85(1-13i)}{(1+13i)(1-13i)} = \frac{85 - (85)13i}{170} = \frac{1}{2} - \frac{13}{2}i.$$

Choosing a close $(p+qi) \in \mathbb{Z}[i]$ we get $(p+qi) = 1-7i$. Finding the product $(1+13i)(1-7i) = 92+6i$. Solving for $\gamma = 85 - (92+6i) = -7-6i$. Double checking norms we get, $N(-7-6i) = 7^2 + 6^2 = 85 \leq 85 = \frac{1}{2}170N(1+13i)$. So our first step looks like,

$$85 = (1-7i)(1+13i) + (-7-6i).$$

Computing the next step we first check the quotient,

$$\frac{1+13i}{-7-6i} = \frac{(1+13i)(-7+6i)}{(-7-6i)(-7+6i)} = \frac{-85-85i}{85} = -1-i.$$

Which gives us our final step,

$$1+13i = (-7-6i)(-1-i) + 0.$$

Therefore the greatest common divisor for 85 and $1+13i$ is $(-7-6i)$.

Applying the Euclidean algorithm for $47-13i$ and $53+56i$, first we compute the following quotient,

$$\frac{53+56i}{47-13i} = \frac{(53+56i)(47+13i)}{(47-13i)(47+13i)} = \frac{1763+3321i}{2378} = \frac{43}{58} + \frac{81}{58}i$$

Choosing a close $(p+qi) \in \mathbb{Z}[i]$ we get $(p+qi) = 1+i$. Finding the product $(47-13i)(1+i) = 60+34i$. Solving for $\gamma = (53+56i) - (60+34i) = -7+22i$. Double checking norma we get, $N(-7+22i) = 533 \leq 1189 = \frac{2378}{2} = \frac{1}{2}N(47-13i)$. So our first step looks like,

$$53+56i = (1+i)(47-13i) + (-7+22i).$$

Computing the next step we first check the quotient,

$$\frac{47 - 13i}{-7 + 22i} = \frac{(47 - 13i)(-7 - 22i)}{(-7 + 22i)(-7 - 22i)} = \frac{-615 - 943i}{533} = -\frac{15}{13} - \frac{23}{13}i.$$

Choosing a close $(p + qi) \in \mathbb{Z}[i]$ we get $(p + qi) = -1 - 2i$. Finding the product $(-7 + 22i)(-1 - 2i) = 51 - 8i$. Computing $\gamma = (47 - 13i) - (51 - 8i) = -4 - 5i$. So our next step is

$$47 - 13i = (-1 - 2i)(-7 + 22i) + (-4 - 5i).$$

For the next step we compute the following quotient,

$$\frac{-7 + 22i}{-4 - 5i} = \frac{(-7 + 22i)(-4 + 5i)}{(-4 - 5i)(-4 + 5i)} = \frac{-82 - 123i}{41} = -2 - 3i.$$

Which gives us our final step,

$$-7 + 22i = (-2 - 3i)(-4 - 5i) + 0.$$

Therefore greatest common divisor for $47 - 13i$ and $53 + 56i$ is $(-4 - 5i)$.

□

Section 8.2

Exercise (8.2.1). Prove that in an Principal Ideal Domain R , two ideals (a) and (b) are comaximal if and only if a greatest common divisor of a and b is 1.

Proof: Suppose R is a Principal Ideal Domain and two ideal (a) and (b) are comaximal. By the definition of comaximal we know that $(a) + (b) = R$. By Proposition 6 we know that if d is a generator for the principal ideal generated by a and b then $d = \gcd(a, b)$. Therefore $(a), (b) \subseteq (d)$ and by (2) of Proposition 6, $(a) + (b) \subseteq (d)$. Since (a) and (b) are comaximal we know that $(a) + (b) = R$ therefore $(d) = R$. Thus $d = 1$ and therefore $\gcd(a, b) = 1$.

□

Exercise (8.2.3). Prove that a quotient of a *P.I.D* by a prime ideal is again a *P.I.D*.

Proof: Suppose R is a *P.I.D* and P is a prime ideal. If P non zero then by Proposition 7 we know that P is maximal in R . By Proposition 12 of Section 7.4, since R is a *P.I.D* it is therefore commutative and thus if P is maximal R/P is a field. Thus R/P has only two ideals 0 and R/P which are both principal, thus R/P is a *P.I.D*. $\square$